Machine Learning for finance

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Outline

1. Introduction
2. Scoring
3. Pecularities of Machine Learning for finance

1 Introduction

1. Introduction

- Statistics vs Data science
- New typologies of data : tables, texts, images, videos ...
- New forms for datasets: structured or not, graphs, labelled or not, etc.
- New ways of obtaining predictive power from data
- Fields of application : health, insurance, finance, sport, education, medicine, economy, marketing, ...

1. Introduction

- Features $X \in \mathbb{R}^d$, $d >> 1$
- Label $Y \in \mathbb{R}$, or $Y \in \{0, 1\}$, or even take a finite number of values
- Data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$
- traditionally, the data entries are i.i.d.
- Objective : predict Y for a new observation X
- We build more an algorithm than a model
- We want to make predictions but may end up with a black box

- Supervised learning: we observe (X_i, Y_i)
- Unsupervised learning: we observe only X_i (no label)
- Semi-supervised learning: data are partly labelled
- Online learning: Some new data comes in regularly
- Active learning: I have an influence on the distribution of the new data coming in

1. Introduction

- Data set made of observations $(X_i, Y_i)_{1 \leq i \leq n}$ which are i.i.d. (w.l.o.g., i.e. to simplify) and have an unknown distribution $\mathbb{P}$.
- $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$.
- We look towards a function of X to predict Y, i.e. a function $g : \mathcal{X} \to \mathcal{Y}$. The predictor of Y is $g(X)$.
- A learning set $(Z_1, \ldots, Z_n)$
- A learning algorithm is a map which associates a prediction function to a dataset:

$$(Z_1, \ldots, Z_n) \in \mathcal{D}_n \longrightarrow \hat{g}_n \in \mathcal{F}(\mathcal{X}, \mathcal{Y}).$$

- We look towards the best prediction function for a given criterion.

1. Introduction

- The Loss function measures the cost of mislabeling a data point:
$$\ell : (y, y') \mapsto \ell(y, y') \in \mathbb{R}$$
- Example 1. $\ell : (y, y') \mapsto 1(y \neq y)$ (binary classification)
- Example 2. $\ell : (y, y') \mapsto |y - y'|^2$ (squared error)
- Example 3. $\ell : (y, y') \mapsto \max\left(0, 1 - yy'\right)$ (hinge loss, $y \in \{1, -1\}$)
- The cost should be designed depending on the application of interest.
- The risk of a prediction function is measured by
$$R_{\mathbb{P}}(g) := \mathbb{E}_{\mathbb{P}}\left[\ell(g(X), Y)\right]$$
- The risk measures the "average" error of a candidate prediction function g.

1. Introduction

- We look for the minimizer $g^*_{\mathbb{P}}$ of the Risk function, i.e.

$$g^*_{\mathbb{P}} = \arg\min_g R_{\mathbb{P}}(g)$$

- A predictor $g^*_{\mathbb{P}}$ is called "oracle" or "Bayes predictor" because it is an unreachable best target.
- It is unreachable as long as $\mathbb{P}$ is unknown
- In practice, focus on the empirical risk

$$\hat{R}_n(g) := \frac{1}{n}\sum_{i=1}^{n} \ell(g(X_i), Y_i)$$

- The oracle for the linear regression criterion is

$$\eta^*_{\mathbb{P}}(x) := \mathbb{E}_{\mathbb{P}}\big[Y|X = x\big]$$

The conditional expectation is the "best approximation" of Y with the information induced by X.

Proof:

$$R_{\mathbb{P}}(g) := \mathbb{E}_{\mathbb{P}}\big[(g(X) - Y)^2\big] = \mathbb{E}_{\mathbb{P}}\big[\mathbb{E}[(g(X) - Y)^2|X]\big]$$

For any value x,

$$\arg\min_g \mathbb{E}[(g - Y)^2|X = x] = \mathbb{E}[Y|X = x]. \ \square$$

- The oracle for binary classification (0 and 1) is

$$g^*_{\mathbb{P}}(x) := 1\big(\mathbb{E}_{\mathbb{P}}\big[Y|X = x\big] \geq 1/2\big)$$

Proof: for any binary classifier $g(X)$, we have

$$\ell\big(g(X), Y\big) = Y.1\big(g(X) \leq 1/2\big) + (1 - Y).1\big(g(X) > 1/2\big).$$

$$\Rightarrow R_{\mathbb{P}}(g) := \mathbb{E}_{\mathbb{P}}\Big[\mathbb{E}[Y|X]1\big(g(X) \leq 1/2\big) + (1 - \mathbb{E}[Y|X])1\big(g(X) > 1/2\big)\Big]$$
$$\geq \quad \mathbb{E}_{\mathbb{P}}\big[\min(E[Y|X], 1 - E[Y|X])\big]$$

But, when $g^*(x) = 1\big(E[Y|X = x] \geq 1/2\big)$, the lower bound is reached. $\square$

- The oracle when $Y \in \{1, \ldots, p\}$ and the 0-1 loss is

$$g^*_{\mathbb{P}}(x) := \arg\max_k \mathbb{P}\big(Y = k | X = x\big)$$

- Connection between the Errors in terms of risk : for a classifier $g = 1(\eta \geq 1/2)$, we have

$$0 \leq R^{cl}_{\mathbb{P}}(g) - R^{cl}_{\mathbb{P}}(g^*_{\mathbb{P}}) \leq 2\sqrt{R^{reg}_{\mathbb{P}}(\eta) - R^{reg}_{\mathbb{P}}(\eta^*_{\mathbb{P}})}$$

1. Introduction

- Theoretical results on the convergence of the algorithm, when the number of points in the training dataset converges to ∞.
- Algorithm/regressor/predictor: $\hat{g}_n$, a map of $(Z_1, \ldots, Z_n)$.
- The algorithm is consistent with respect to $\mathbb{P}$ if and only if

$$\mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(\hat{g}_n)\right] \underset{n\to\infty}{\longrightarrow} \mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(g^*_{\mathbb{P}})\right].$$

- The algorithm is consistent with respect to a set of probabilities $\mathcal{P}$ if and only if

$$\mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(\hat{g}_n)\right] \underset{n\to\infty}{\longrightarrow} \mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(g^*_{\mathbb{P}})\right] \quad \forall \mathbb{P} \in \mathcal{P}.$$

- We can never get $\lim_n \sup_{\mathbb{P}} |\mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(\hat{g}_n)\right] - \mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(g^*_{\mathbb{P}})\right]| = 0$.
- But we may be able to derive

$$\sup_{\mathbb{P}} \lim_n |\mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(\hat{g}_n)\right] - \mathbb{E}_{\mathbb{P}}\left[R_{\mathbb{P}}(g^*_{\mathbb{P}})\right]| = 0$$

1. Introduction

- The (theoretical) risk is computed in practice via its empirical counterpart

$$\hat{R}_n(g) := \frac{1}{n}\sum_{i=1}^{n} \ell(g(X_i), Y_i)$$

- With i.i.d data, the Law of Large Numbers (resp. Central Limit Theorem) provides the consistency (resp. asymptotic normality) of $\hat{R}_n(g)$ (resp. $\sqrt{n}\big(\hat{R}_n(g) - R_{\mathbb{P}}(g)\big)$).
- Central question in machine learning : what is the set of admissible prediction function ?
- We restrict to a subset of possible predictors: $g \in \mathcal{G}$.
- If $\mathcal{G}$ is too large, risk of over-fitting, lack of robustness.
- If $\mathcal{G}$ is too small, risk of not allowing enough predicting power.

Bias-variance trade-off

- We need to measure the complexity of $\mathcal{G}$
- For a given admissible set of predictors, we define

$$g^*_{\mathbb{P},\mathcal{G}} = \arg\min_{g\in\mathcal{G}} R_{\mathbb{P}}(g), \quad \hat{g}_{n,\mathcal{G}} = \arg\min_{g\in\mathcal{G}} \hat{R}_n(g).$$

- $R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) \geq R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) \geq R_{\mathbb{P}}(g^*_{\mathbb{P}})$
- We decompose the excess of risk in the following way

$$\begin{aligned} R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P}}) &= R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) \ \text{(stochastic error/variance} \\ &+ \quad R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P}}) \ \text{(systematic error/bias)} \end{aligned}$$

- The choice of $\mathcal{G}$ should provide a balance between these terms
- The variance is controlled by the following

$$R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) \leq 2 \sup_{g\in\mathcal{G}} |R_{\mathbb{P}}(g) - \hat{R}_n(g)|$$

Performance of an algorithm: a simple case

- Require the use of concentration (exponential) inequalities
- Hoeffding inequality: let $(V_1, \ldots, V_n)$ be i.i.d. $\sim V$, $a_i \leq V_i \leq b_i$, and $\overline{V}_n := n^{-1} \sum_{i=1}^n V_i$.

$$\mathbb{P}\Big(\hat{V}_n - \mathbb{E}[V] > t\Big) \leq \exp\Big(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big), \ \forall t > 0$$

- We deduce a control of the error in terms of risk. If the loss function satisfies $a \leq \ell \leq b$ and the cardinal of $\mathcal{G}$ is finite: for every $\varepsilon > 0$,

$$\mathbb{P}\Big(R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) \leq (b-a)\sqrt{\frac{2\ln(2|\mathcal{G}|/\varepsilon)}{n}}\Big) \geq 1 - \varepsilon$$

Performance of an algorithm: a simple case

- More generally, require to create a measure of complexity for $\mathcal{G}$: the Vapnik-Chervonenkis dimension $V(\mathcal{G})$ of $\mathcal{G}$.
- for binary classifiers,

$$\mathbb{P}\Big(R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) - R_{\mathbb{P}}(g^*_{\mathbb{P},\mathcal{G}}) \leq \sqrt{\frac{2V(\mathcal{G})\ln(4(2n+1)/\varepsilon)}{n}}\Big) \geq 1-\varepsilon$$

- Oracle inequalities take the form

$$\mathbb{P}\Big(R_{\mathbb{P}}(\hat{g}_{n,\mathcal{G}}) \leq A \inf_{g\in\mathcal{G}} R_{\mathbb{P}}(g) + B\sqrt{\frac{\ln(\mathcal{C}(\mathcal{G})/\varepsilon)}{n}}\Big) \geq 1-\varepsilon$$

for some complexity measure $\mathcal{C}(\mathcal{G})$ (depends on n). The latter inequality is sharp if $A = 1$.

ML methods and their "regularizers"

Function class $\mathcal{F}$ (and its parameterization)	Regularizer $R(f)$
Global/parametric predictors	
Linear $\beta' x$ (and generalizations)	Subset selection $\|\beta\|_0 = \sum_{j=1}^k \mathbf{1}_{\beta_j \neq 0}$
	LASSO $\|\beta\|_1 = \sum_{j=1}^k \|\beta_j\|$
	Ridge $\|\beta\|_2^2 = \sum_{j=1}^k \beta_j^2$
	Elastic net $\alpha\|\beta\|_1 + (1-\alpha)\|\beta\|_2^2$
Local/non-parametric predictors	
Decision/regression trees	Depth, number of nodes/leaves, minimal leaf size, information gain at splits
Random forest (linear combination of trees)	Number of trees, number of variables used in each tree, size of bootstrap sample, complexity of trees (see above)
Nearest neighbors	Number of neighbors
Kernel regression	Kernel bandwidth
Mixed predictors	
Deep learning, neural nets, convolutional neural networks	Number of levels, number of neurons per level, connectivity between neurons
Splines	Number of knots, order
Combined predictors	
Bagging: unweighted average of predictors from bootstrap draws	Number of draws, size of bootstrap samples (and individual regularization parameters)
Boosting: linear combination of predictions of residual	Learning rate, number of iterations (and individual regularization parameters)
Ensemble: weighted combination of different predictors	Ensemble weights (and individual regularization parameters)

2 Scoring

The goal

Discriminate between several categories of observations, given some covariates.

Formally: realizations of (X, Y), where

- $X \in \mathbb{R}^p$, with continuous and/or discrete components,
- a discrete explained variable $Y \in \{0, \ldots, q-1\}$

$\Rightarrow$ classification!

We are particularly interested in the binary case ($q = 2$).

Standard examples in finance

- credit assessment for a loan/bond: $Y = 1$ iff the potential borrower will be "good" for the lender ("no credit event" is likely). $X =$ vector of individual characteristics, past repayment records...(c.f. FICO scores in the US)
- Y is the rating of a firm (credit standing, Basel 2 IRB approach); $X =$ financial ratios, past performances, macro-economic forecasts etc.
- Y is a grade for the purpose of customer segmentation (profitability) and/or marketing.
- $Y = 1$ when the financial asset has better returns than "the market" (a positive α). $X =$ technical indicators, past returns/vol., e.g.

A bit of decision theory

Predictor $\hat{Y} = \hat{Y}(X)$.

The final objective is to predict a discrete variable by another discrete variable.

Nonetheless, this is done most often through

- continuous predictors $\hat{S}(X) \in \mathbb{R}$
- and some thresholds: $(\hat{Y}(x) = k) = (\hat{S}(x) \in [a_k, a_{k+1}))$, where

$$a_0 = -\infty < a_1 < \ldots < a_{q-1} < a_q = +\infty$$

In this course, $\hat{Y}(X)$ will be discrete or continuous, depending on the context.

A bit of decision theory

Loss function L and Expected Prediction Error

$$EPE = \mathbb{E}\big[L(Y, \hat{Y}(X))\big] = \mathbb{E}_X\Big[\sum_{k=0}^{q-1} \mathbb{E}\big[L(k, \hat{Y}(X))\big]\mathbb{P}(Y = k|X)\Big]$$

An optimal classifier is map $\hat{F} : X \mapsto \{0, \ldots, q-1\}$ that minimizes EPE pointwise, i.e.

$$\hat{F}(x) = \arg\min_{\ell \in \{0,\ldots,q-1\}} \sum_{k=0}^{q-1} \mathbb{E}\big[L(k, \ell)\big]\mathbb{P}(Y = k|X = x), \ \forall x.$$

If $L(k, \ell) = 1(k \neq \ell)$,

$$\hat{F}(x) = \arg\min_{\ell \in \{0,\ldots,q-1\}} \big(1 - \mathbb{P}(Y = \ell|X = x)\big), \ \forall x.$$

Bayes classifier:

$$\hat{F}(x) = \ell \Leftrightarrow \mathbb{P}(Y = \ell|X = x) = \max_k \mathbb{P}(Y = k|X = x).$$

A bit of decision theory

$\Rightarrow$ we need a model for the calculation of $\mathbb{P}(Y = k|X = x)$ for any k and x (the "class posterior" probabilities).

Denote $\pi_j = \mathbb{P}(Y = j)$ (priors) and $f_j(x) = \mathbb{P}(X = x|Y = j)$, $\forall j$.

$$\mathbb{P}(Y = k|X = x) \overset{Bayes}{=} \frac{f_k(x)\pi_k}{\sum_{j=0}^{q-1} f_j(x)\pi_j} \propto f_k(x)\pi_k$$

$\Rightarrow$ two "equivalent" ways of thinking:

(i) select the optimal Bayes classifier by comparing the $\mathbb{P}(Y = k|X = x)$, for every k: limited dependent variable models, duration models, trees, neural networks, etc.

(ii) select k s.t. $f_k(x)\pi_k \geq f_j(x)\pi_j$ for every j: "discriminant analysis". When $q = 2$, this means select $Y = 1$ when $f_1(x)/f_0(x) > t$ for some threshold t.

Discriminant Analysis: choice of the threshold

The threshold can be calibrated following an economic reasoning.

Assume $q = 2$, and a score for loan allocation.

- $g =$ the average gain of a good decision (accept the loan application for a good borrower).

  Probability of occurrence $= \mathbb{P}(\hat{Y}(X) = 1, Y = 1)$.
- $c_0 =$ cost of allocating a loan to a "bad" borrower.
  Probability of occurrence $= \mathbb{P}(\hat{Y}(X) = 1, Y = 0)$.
- $c_1 =$ cost of refusing a loan to a "good" borrower.

  Probability of occurrence $= \mathbb{P}(\hat{Y}(X) = 0, Y = 1)$.

Discriminant Analysis: choice of the threshold

The loss function is then

$$\begin{aligned} Loss &= c_0 \mathbb{P}(\hat{Y}(X) = 1, Y = 0) + c_1 \mathbb{P}(\hat{Y}(X) = 0, Y = 1) \\ &- \quad g \mathbb{P}(\hat{Y}(X) = 1, Y = 1) \end{aligned}$$

Denote by $\mathcal{A} = \{x | \hat{Y}(x) = 1\} \subset \mathbb{R}^p$ the acceptance area.

$$\begin{aligned} Loss &= c_0 \mathbb{P}(\hat{Y}(X) = 1 | Y = 0)\pi_0 + c_1 \mathbb{P}(\hat{Y}(X) = 0 | Y = 1)\pi_1 \\ &- \quad g \mathbb{P}(\hat{Y}(X) = 1 | Y = 1)\pi_1 \\ &= \quad c_1 \pi_1 - \int_{\mathcal{A}} \left\{ (g + c_1)\pi_1 f_1(x) - c_0 \pi_0 f_0(x) \right\} dx \end{aligned}$$

Minimize the loss w.r.t. $\mathcal{A}$:

$$\mathcal{A}^*(x) := \{x \,|\, \frac{f_1(x)}{f_0(x)} \geq \frac{c_0 \pi_0}{(g + c_1)\pi_1} \}$$

Discriminant Analysis: choice of the threshold

Bayes formula:

$$\mathbb{P}(Y = 1|X = x) = \frac{\pi_1 f_1(x)}{\pi_0 f_0(x) + \pi_1 f_1(x)} = \frac{\pi_1}{\pi_0 f_0(x)/f_1(x) + \pi_1}$$

$$\Rightarrow \mathcal{A}^*(x) := \left\{x \,|\, \mathbb{P}(Y = 1|X = x) \geq \frac{c_0}{g + c_0 + c_1}\right\}$$

The relevant probability threshold is not always 1/2 !

To do: model specification + estimation of every class densities f_k, $k \in \{0, 1, \ldots, q-1\}$.

$\Rightarrow$ q parametric/nonparametric/semi-parametric models

Most of the time, all the f_k belong to the same family/class of models.

Inference on convenient sub-samples: with a N sample size,

- $\hat{\pi}_k := N_k/N$, $N_k = \sum_{i=1}^{N} 1(Y_i = k)$ (when the sample is not biased),
- or even $\pi_k = 1/q$ (no prior).

Discriminant Analysis: the Gaussian case

Assumption: for every k, f_k is the density of a random vector whose law is $\mathcal{N}(\mu_k, \Sigma_k)$.

Linear Discriminant Analysis (LDA) assumption: $\Sigma_k = \Sigma$ for any k.

$$\begin{aligned}\ln\left(\frac{\mathbb{P}(Y=k|X=x)}{\mathbb{P}(Y=\ell|X=x)}\right) &= \ln\frac{f_k(x)}{f_\ell(x)} + \ln\frac{\pi_k}{\pi_\ell} \\ &= \ln\frac{\pi_k}{\pi_\ell} + x'\Sigma^{-1}(\mu_k-\mu_\ell) - \frac{1}{2}\mu_k'\Sigma^{-1}\mu_k + \frac{1}{2}\mu_\ell'\Sigma^{-1}\mu_\ell\end{aligned}$$

$\Rightarrow$ the decision boundary is the hyperplan $x'\Sigma^{-1}(\mu_k-\mu_\ell) = c$, for some constant c.

Estimation of μ_k by $\hat{\mu}_k := \sum_{i=1}^N 1(Y_i = k)X_i/N_k$.

Estimation of Σ by

$$\hat{\Sigma} := \sum_{i=1}^N \sum_{k=0}^{q-1} 1(Y_i = k)(X_i - \hat{\mu}_k)(X_i - \hat{\mu}_k)'/N$$

Discriminant Analysis; the Gaussian case

If the covariance matrices Σ_k are different: Quadratic Discriminant Analysis (QDA)

Introduce the discriminant functions

$$\delta_k(x) := -\frac{1}{2}\ln|\Sigma_k| - \frac{1}{2}(x-\mu_k)'\Sigma_k^{-1}(x-\mu_k) + \ln\pi_k.$$

$$\{x : \mathbb{P}(x : Y = k|X = x) > \mathbb{P}(Y = \ell|X = x)\} = \{x : \delta_k(x) > \delta_\ell(x)\}$$

The separating boundaries $\{x : \delta_k(x) = \delta_\ell(x)\}$ are quadratic.

This necessitates the estimation of every Σ_k by

$$\hat{\Sigma}_k := \sum_{i=1}^{N} 1(Y_i = k)(X_i - \hat{\mu}_k)(X_i - \hat{\mu}_k)'/N_k$$

Discriminant Analysis; the Gaussian case

Variations: Regularized Discriminant Analysis.

Idem, but adding a regularization stage concerning covariances.

For some $\gamma \in [0, 1]$, set

- $\hat{\Sigma}(\gamma) := \gamma\hat{\Sigma} + (1-\gamma)\sigma^2 \mathrm{Id} \sim$ Ridge regression; relevant with the dimension of Σ is large;
- $\hat{\Sigma}_k(\gamma) := \gamma\hat{\Sigma}_k + (1-\gamma)\hat{\Sigma}$ more stability when q is large.

Discriminant Analysis: another view of LDA

When $q = 2$, LDA $\equiv$ separate into two groups.

Ronald Fisher (1936): find a linear combination $Z := w'X$ that best separates the two groups $Y = 1$ and $Y = 0$.

Proposal:

$$w^* := \arg\max_w \frac{\{\mathbb{E}[Z|Y=1] - \mathbb{E}[Z|Y=0]\}^2}{Var(Z)}, \text{ or}$$

$$w^* := \arg\max_w \frac{w'(\mu_1 - \mu_0)(\mu_1 - \mu_0)'w}{w'\Sigma w} \propto \Sigma^{-1}(\mu_1 - \mu_0)$$

$\Rightarrow$ a separating hyperplane $\{x | x'\Sigma^{-1}(\mu_1 - \mu_0) = c\}$

...without any Gaussian assumption.

Discriminant Analysis: another view of LDA

With different matrices Σ_k per sub-group, set

- the "within-class" variance $n\hat{\Sigma}_W := n_1\hat{\Sigma}_1 + n_0\hat{\Sigma}_0$,
- the "between-class" variance $n\hat{\Sigma}_B := n_1(\hat{\mu}_1 - \hat{\mu})(\hat{\mu}_1 - \hat{\mu})' + n_0(\hat{\mu}_0 - \hat{\mu})(\hat{\mu}_0 - \hat{\mu})'$

$$\hat{w}^{**} := \arg\max_w \frac{w'\hat{\Sigma}_B w}{w'\hat{\Sigma}_W w}$$

The separating hyperplane has the maximum between-class variance relative to the within-class variance.

Discriminant Analysis: regression and LDA

Another point of view concerning LDA: find $\mathbf{w}$ such that

$$Y = w_0 + X'\mathbf{w} + \varepsilon, \ \mathbb{E}[\varepsilon|X] = 0, \text{ or}$$

$$P(Y = 1|X = x) = w_0 + x'\mathbf{w}.$$

Here, note that $Y \in \{0, 1\}$ and $X \in \mathbb{R}^p$...rather non standard !

Order the observations s.t. $Y_i = 1$ iff $i \leq n_1$.

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \sum_{i=1}^{n_1} (1 - w_0 - \mathbf{w}'X_i)^2 + \sum_{i=n_1+1}^{n} (w_0 + \mathbf{w}'X_i)^2$$

Discriminant Analysis: regression and LDA

Notations: $\mathbf{X}_1' = [X_1, \ldots, X_{n_1}]$, $\mathbf{X}_0' = [X_{n_1+1}, \ldots, X_n]$, $\mathbf{e}_1$ (resp. $\mathbf{e}_0$) column vector of ones with size n_1 (resp. $n_0 := n - n_1$).

$$\begin{bmatrix} \mathbf{e}_1 & \mathbf{X}_1 \\ \mathbf{e}_0 & \mathbf{X}_0 \end{bmatrix} =: \mathbf{Z}, \quad \begin{bmatrix} w_0 \\ \mathbf{w} \end{bmatrix} =: \vec{\mathbf{w}}, \quad \begin{bmatrix} \mathbf{e}_1 \\ 0 \end{bmatrix} =: \mathbf{b}$$

Discriminant Analysis: regression and LDA

$$\hat{\mathbf{w}} = \arg\min_{\vec{\mathbf{w}}} (\mathbf{b} - \mathbf{Z}\vec{\mathbf{w}})'(\mathbf{b} - \mathbf{Z}\vec{\mathbf{w}})$$

First-order conditions: $\mathbf{Z}'\mathbf{Z}\vec{\mathbf{w}} = \mathbf{Z}'\mathbf{b}$.

Block-matrix calculations, and define $\hat{\Sigma}$ by

$$\mathbf{X}_1'\mathbf{X}_1 + \mathbf{X}_0'\mathbf{X}_0 = n\hat{\Sigma} + n_1\hat{\mu}_1\hat{\mu}_1' + n_0\hat{\mu}_0\hat{\mu}_0'.$$

After some algebraic manipulations, we get

$$\hat{\mathbf{w}} \propto \hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_0)$$

$\Rightarrow$ again the separating hyperplane $\{x | x'\hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_0) = c\}$ ...still without any Gaussian assumption.

Other models of conditional densities of X given Y:

- mixture of Gaussian laws $\rightarrow$ similar to QDA.
- nonparametric density estimators: Nadarya-Watson kernel, nearest-neighbors, local-linear, wavelets, etc.

$$\hat{f}_k(x) := \frac{1}{n_k h^p} \sum_{i=1}^{n} 1(Y_i = k) K\Big(\frac{x - X_i}{h_n}\Big),$$

for some kernel $K : \mathbb{R}^p \rightarrow \mathbb{R}$, $\int K = 1$, and a "bandwidth" sequence (h_n) of positive numbers s.t. $h_n \rightarrow 0$, $nh_n^p \rightarrow +\infty$. Well-suited when p is small (otherwise: curse of dimension).

Probabilistic models of Y given X

The alternative approach: model for $\mathcal{L}(Y|X)$ instead of $\mathcal{L}(X|Y)$.

The target: evaluate $\mathbb{P}(Y = k|X = x)$ for any $k \in \{0, \ldots, q-1\}$ and any $x \Rightarrow$ classification.

Many methods available...known for a long time or renewed by the Machine Learning revolution.

Logistic regression

q classes and $k = 0$ is the reference one.

$$\ln \frac{\mathbb{P}(Y = k|X = x)}{\mathbb{P}(Y = 0|X = x)} = \beta_{0,k} + x'\beta_k, \; k = 1, \ldots, q-1.$$

$$\Rightarrow \mathbb{P}(Y = k|X = x) = \frac{\exp(\beta_{0,k} + x'\beta_k)}{1 + \sum_{\ell=1}^{q-1} \exp(\beta_{0,\ell} + x'\beta_\ell)}$$

When $q = 2$, the usual "Logit model".

Unkown parameter $\theta := (\beta_{0,1}, \beta_1, \ldots, \beta_{0,q-1}, \beta_{q-1})$.

Logistic regression: inference

By Maximum likelihood:

$$\begin{aligned}\mathrm{Lik}_n(\theta) &= \sum_{i=1}^{n}\sum_{k=0}^{q-1} 1(Y_i = k)\ln \mathbb{P}(Y = k|X = X_i) \\ &= \sum_{i=1}^{n}\left\{\beta_{0,Y_i} + X_i'\beta_{Y_i} - \ln\left(1 + \sum_{\ell=1}^{q-1}\exp(\beta_{0,\ell} + X_i'\beta_\ell)\right)\right\}\end{aligned}$$

$\Rightarrow$ usual MLE inference, consistency, asymptotic normality, asymptotic efficiency, etc.

Logistic regression: the bivariate case

Assume only two classes 0 and 1 ($q = 2$). The parameter is $\beta := [\beta_{0,1}, \beta_1]$; include the constant 1 inside X.

$$\begin{aligned}
\mathrm{Lik}_n(\beta) &= \sum_{i=1}^{n} \Big\{ Y_i \ln \mathbb{P}(Y = 1 | X = X_i) \\
&+ \quad (1 - Y_i) \ln \big(1 - \mathbb{P}(Y = 1 | X = X_i)\big) \Big\} \\
&= \quad \sum_{i=1}^{n} \Big\{ Y_i X_i' \beta - \ln \big(1 + \exp(X_i' \beta)\big) \Big\}
\end{aligned}$$

$$\nabla_\beta \mathrm{Lik}_n(\beta) = \sum_{i=1}^{n} \Big\{ X_i \big(Y_i - \mathbb{P}(Y = 1 | X_i, \beta)\big) \Big\}$$

Nonlinear equations $\Rightarrow$ numerical procedures, as coordinate-descent.

Logit models and LDA

Assume we satisfy the LDA assumption: f_k is the density of a $\mathcal{N}(\mu_k, \Sigma)$, for $k \in \{0, 1\}$.

$$\begin{aligned} \ln\left(\frac{\mathbb{P}(Y=1|x)}{1-\mathbb{P}(Y=1|x)}\right) &= \ln\left(\frac{\pi_1 f_1(x)}{\pi_0 f_0(x)}\right) \\ &= \quad 2x'\Sigma^{-1}(\mu_0 - \mu_1) + Const \end{aligned}$$

$\Rightarrow$ LDA satisfies the Logistic Regression assumption.

The reverse is false: there exist some Logit models that are not given by conditional Gaussian X-laws !

$\Rightarrow$ assuming Gaussian cdfs' for the f_k is generally a strong assumption for only the purpose of classification.

Logit models and high dimension

Sometimes, a large number of explanatory variables: $p >> 1$, $p \sim q$ or even $p >> n$.

$\Rightarrow$ model selection via penalization.

A new "penalized" criterion

$$Loss_n(\theta) := -\mathrm{Lik}_n(\theta) + \lambda\, \mathrm{pen}(\theta),$$

where

- $\mathrm{pen}(\theta)$ enforces sparsity,
- $\lambda = \lambda(n, p)$ is a tuning parameter, to be fixed

Logit models and high dimension

Many penalizations have been proposed.

- Lasso: $\text{pen}(\theta) = \sum_{j=1}^{p} |\theta_j| = \|\theta\|_1$
- elastic net: $\text{pen}(\theta) = \alpha\|\theta\|_1 + (1-\alpha)\|\theta\|_2^2$, $\alpha \in [0,1]$
- Group Lasso: m groups of coefficients, $\theta = [\theta_1, \ldots, \theta_m]$, and

$$\text{pen}(\theta) = \sum_{\ell=1}^{m} \alpha_\ell \|\theta_\ell\|_2$$

- Ordered Lasso: to enforce an implicit ordering among (some of) the coefficients
- Fused Lasso: to enforce an implicit stability among (some of) the coefficients,

+ others: SCAD, MCP, sparse Group Lasso, etc.

Logit models and high dimension: cross-validation

A usual way of calibrating λ: K-fold cross-validation.

1. split the training set into K equal-sized parts;
2. fit the penalized model on $K-1$ subsets of observations successively $\rightarrow$ K predictors $\hat{p}_1, \ldots, \hat{p}_K$;
3. for any $i \in \{1, \ldots, n\}$, the index $\nu(i) \in \{1, \ldots, K\}$ denotes of the subset that contains the observation i. Calculate the cross-validation estimation of the OOS prediction error

$$CV(\lambda) := n^{-1} \sum_{i=1}^{n} Loss\big(Y_i, \hat{p}_{\nu(i)}(X_i)\big).$$

   For example ($q = 2$),

$$Loss\big(Y_i, \hat{p}_{\nu(i)}(X_i)\big) = -Y_i X_i' \hat{\beta}_{\nu(i)} + \big\{1 + \exp(X_i' \hat{\beta}_{\nu(i)})\big\} + \lambda \|\hat{\beta}_{\nu(i)}\|_1.$$

4. an optimal tuning parameter is $\hat{\lambda} = \arg\min_\lambda CV(\lambda)$.

Variations around Logistic regressions

When $q = 2$, for any fixed map $F : \mathbb{R} \to \mathbb{R}$, we could assume

$$\mathbb{P}(Y = 1|X = x) = F(x'\beta)$$

A desirable property: F monotonic!

Ex.: Probit model, where $F = \Phi$.

More generally, it is tempting to assume

$$\mathbb{P}(Y = k|X = x) = \psi_k(\beta_k, x), \; k = 0, \ldots, q-1,$$

for some maps $\psi_k : \Theta \times \mathbb{R}^q \to [0, 1]$.

Warning: we have to satisfy $\sum_{k=0}^{q-1} \mathbb{P}(Y = k|X = x) = 1$ for every x !

Odered Logit/Probit

A common situation with scoring:

- q naturally "ordered" categories (ex: credit standing)
- a single univariate "latent variable" (in general, an index $\beta' X$)
- a link function F (a cdf) and some thresholds $a_1, \ldots, a_{q-1}$. Setting $a_0 = -\infty$, and $a_q = +\infty$, $\forall k \in \{0, 1, \ldots, q-1\}$

$$\mathbb{P}(Y = k | X = x) = F(a_{k+1} + \beta' x) - F(a_k + \beta' x).$$

Ex.: $F = \Phi$, F is Logistic.

C.f. the discrete multinomial choice models (Daniel McFadden, 80s')...when there is a natural Y order (ratings, financial strength, profitability, etc).

k-nearest neighbors

When p is not "too large", a simple classification rule:

- fix a distance d in the X-space $\mathbb{R}^p$, and a number of neighbors k;
- for every i, consider $\mathcal{S}_i$, the k sample points j that are the closest to X_i;
- $\hat{Y}(X_i)$ is given by the "majority rule" inside $\mathcal{S}_i$.

In practical terms:

- k too large $\rightarrow$ high bias, low variance,
- k too small $\rightarrow$ low bias, high variance.

Other probabilistic models for classification

When $Y \in \{0,1\}$ and the decision rule is deduced from $\mathbb{E}[Y|X=x]$: any model for regression can be used for classification!

In ML, the natural loss for binary classification purpose is

$$L(\hat{Y}(x), y) = -y \ln \hat{\mathbb{E}}[Y|X=x] - (1-y) \ln \left(1 - \hat{\mathbb{E}}[Y|X=x]\right)$$

$\Rightarrow$ potential use of

- neural networks: for q categories, $Y \in \{0, 1, \ldots, q-1\} \rightarrow$ the predicted vector is $\tilde{Y} \in \mathbb{R}^q$,

$$\tilde{Y}_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0), \ \forall i \in \{1, \ldots, q\}$$

- wavelets: very flexible discontinuous functions
- additive models: $\mathbf{X}'\beta \rightarrow \sum_{j=1}^{p} f_j(X_j)$.
- kernel ridge regression (RKHS methods), etc.

Classification/regression trees

Objective: build homogeneous cells of individuals obtained by the intersection of several half-spaces (CART)

The predicted class is given by the majority rule in every final cell J:

$$k_J^* = \arg \max_{k \in \{0,\dots,q-1\}} \hat{p}_k, \ \hat{p}_k := \frac{1}{|J|} \sum_{i \in J} 1(Y_i = k)$$

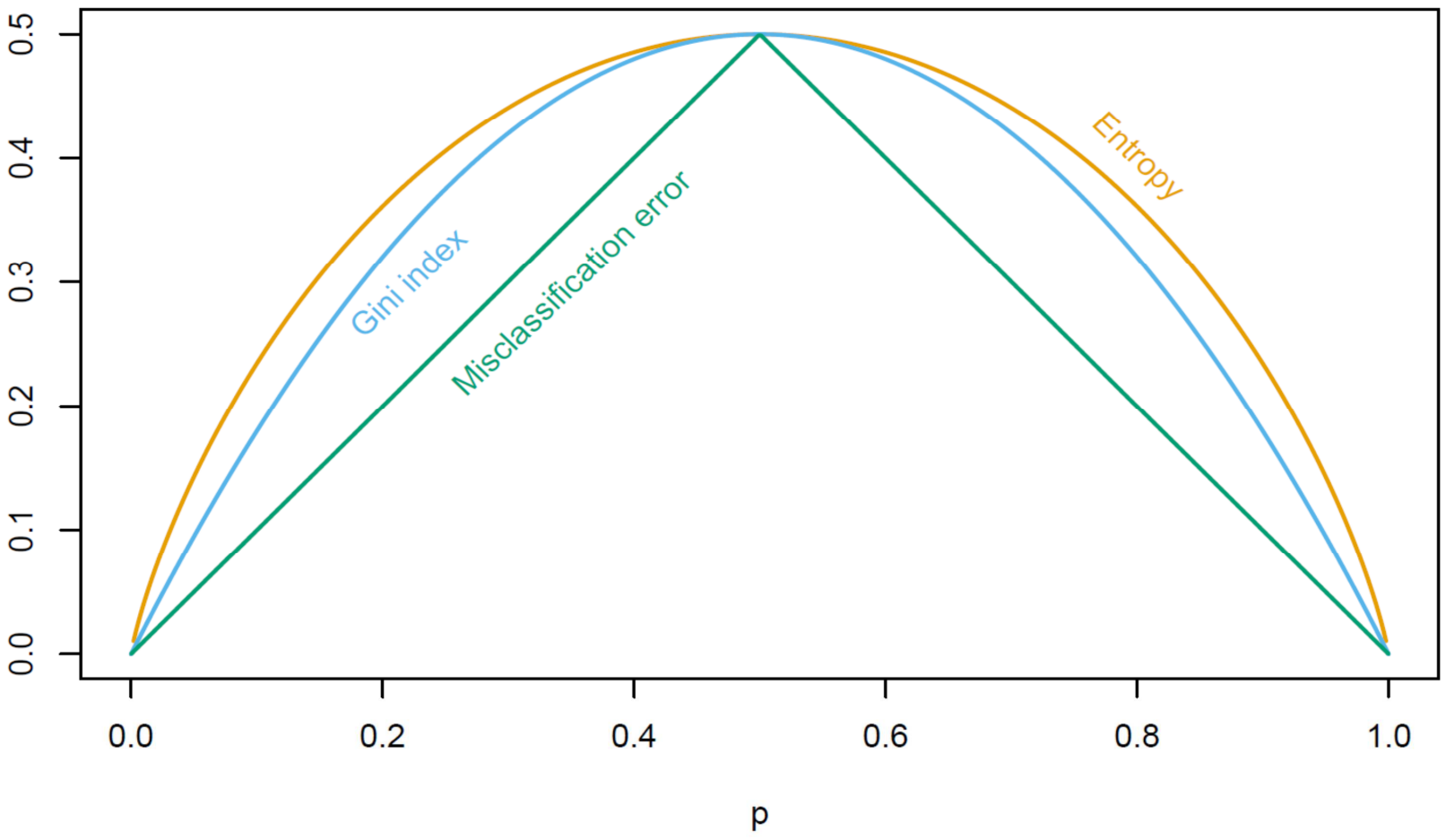
Define an impurity measure $Q(J)$ in any subgroup J of individuals.

- misclassification error rate

$$Q(J) = |J|^{-1} \sum_{i \in J} 1(Y_i \neq k_J^*) = 1 - \hat{p}_{k_J^*}$$

- Gini index $Q(J) = \sum_{k \neq \ell} \hat{p}_k \hat{p}_\ell = \sum_{k=0}^{q-1} \hat{p}_k (1 - \hat{p}_k)$
- entropy $Q(J) = -\sum_{k=0}^{q-1} \hat{p}_k \ln \hat{p}_k$.

Node impurity measures for two-class classification



Classification/regression trees

If the tree $\mathcal{T}$ has M leaves $J_1, \ldots, J_M$ with cardinalities $c_i := |J_i|$, the (penalized) loss function is

$$L(\mathcal{T}) := \sum_{i=1}^{M} c_i Q(J_i) + \lambda M$$

λ measures the trade-off between tree size and goodness-of-fit

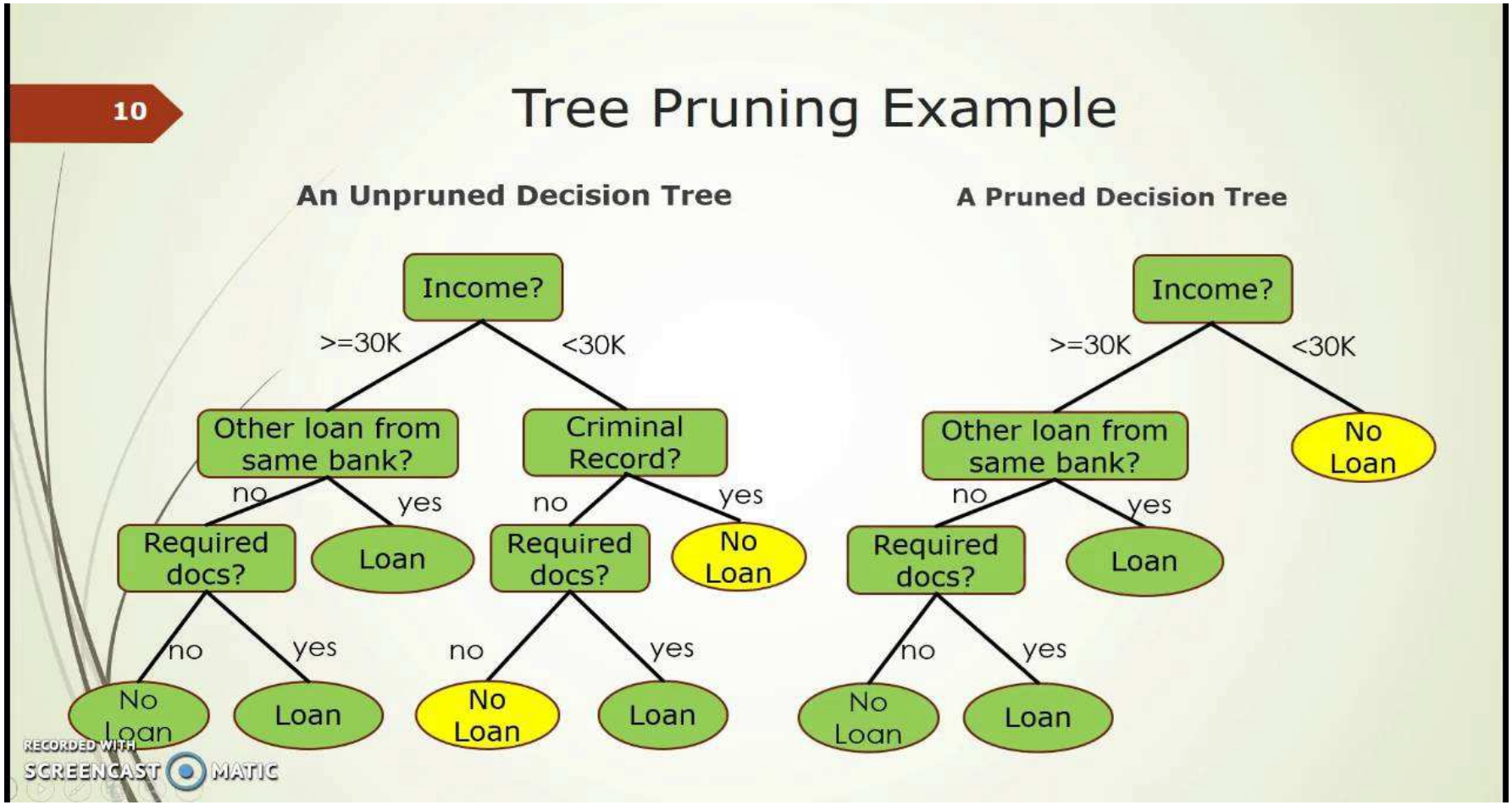
Choose λ by cross-validation.

Classification/regression trees

A usual "greedy" algorithm:

- for every $(j, s) \in \{1, \ldots, p\} \times \mathbb{R}$ split the subsample as $R_{1j} := \{i | X_{ij} \leq s\}$ and $R_{2j} := \{i | X_{ij} > s\}$; calculate $Q(R_{1j})$ and $Q(R_{2j})$.
- find the pair (j^*, s^*) that minimizes $|R_{1j}|Q(R_{1j}) + |R_{2j}|Q(R_{2j})$ (the "splitting rule");
- repeat overs all leaves and increase the size of the tree until an arbitrarily given level is reached (in terms of size, loss, impurity...);
- reduce the size of the tree by "pruning": find a subtree that minimizes $L(\mathcal{T})$.

  Weakest link pruning: successively collapse the internal node that produces the smallest per-node increase of $\sum_{i=1}^{M} c_i Q(J_i)$.


10
Tree Pruning Example
An Unpruned Decision Tree
A Pruned Decision Tree
Income?
>=30K
<30K
Other loan from same bank?
Criminal Record?
no
yes
Required docs?
Loan
No Loan
RECORDED WITH SCREENCAST MATIC

Model averaging: random forests

The idea: average many noisy but approximately unbiased models, and hence reduce the variance ("bagging")

- train B classification trees, based on diverse sub-samples and subsets of $p' < p$ explanatory variables
- the final predictor is the average of all the latter predicted probabilities and/or given by the majority vote among all the trees.

$$\mathbb{P}(\hat{Y}_{RF} = k|X = x) = B^{-1} \sum_{b=1}^{B} \mathbb{P}(\hat{Y}_b = k|X = x), \ \forall k, \text{ or}$$

$$\hat{Y}_{RF}(x) = B^{-1} \sum_{b=1}^{B} \hat{Y}_b(x), \text{ when } p = 2, \text{ or}$$

$\hat{Y}_{RF}(x)$= the most frequently met class among $\hat{Y}_b(x)$.

Duration models/Survival analysis

Relevant when the underlying phenomenon is related to durations/lifetimes:

- duration between inception of a loan and a potential credit event
- lifetimes of companies
- durations between trades/transactions/deals
- date of a potential drawing for a line of credit, etc.

Duration models can play the same role as scoring models: survival/default probabilities, expected remaining lifetimes,

cumulative hazard curves, etc, may be seen as scores.

Duration models

Classical parametric/semiparametric models:

- proportional hazard models
- accelerated time models
- competing risk models

A huge amount of academic and professional literature.

Specialized journals: Biometrics, Biometrika (+ biostatistics/medicine/biology)

Duration models

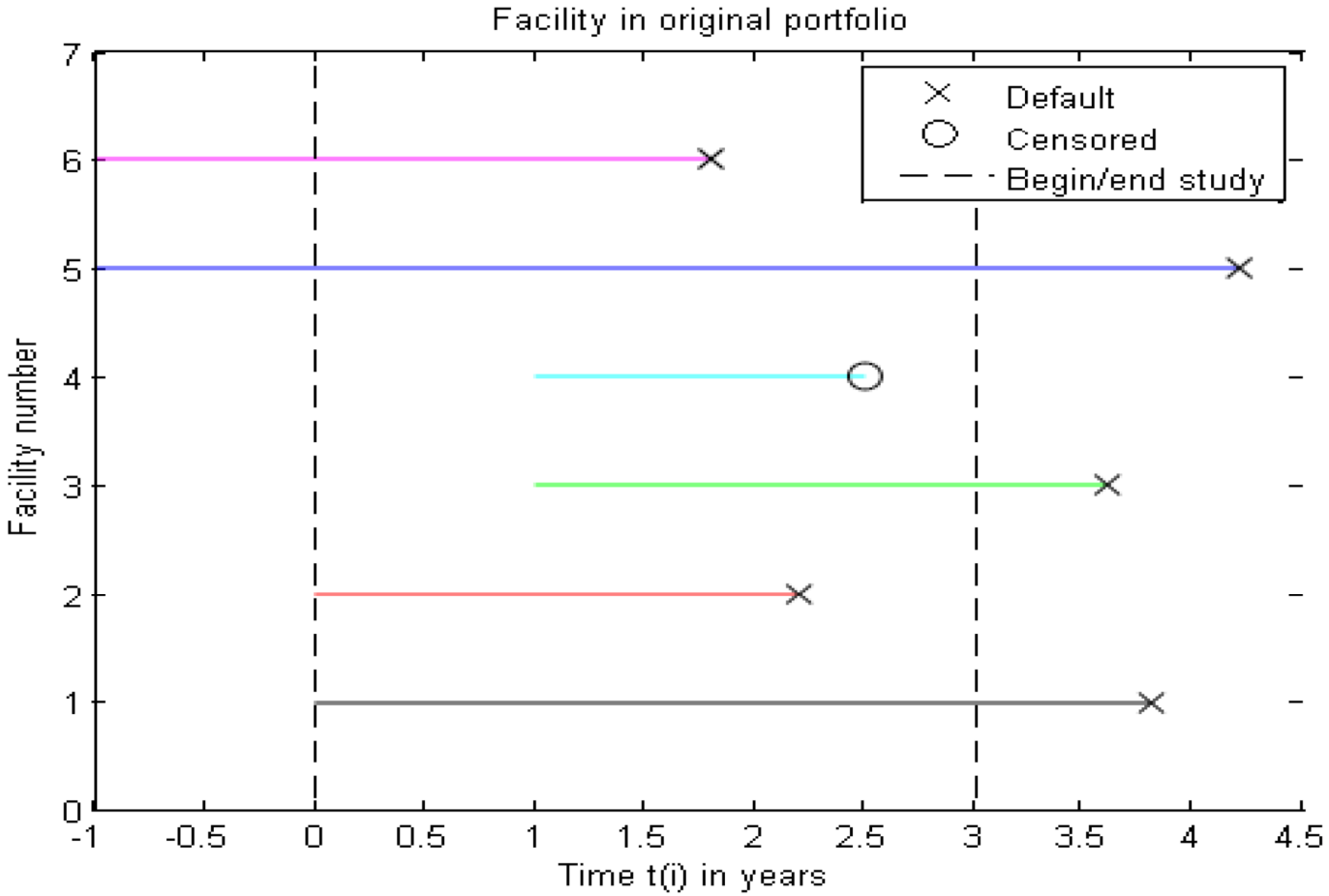
Until now, Machine Learning has not yet created a revolution in survival analysis. Why?

Typical problems in duration models: partial observability of the underlying random variables.

Ex.: censoring, truncation, selection biases, etc.

These are challenging theoretical problems for usual ML techniques (unusual sources of biases), but progresses.

C.f. Sirignano, Sadhwani & Giesecke (2018), Deep Learning for mortgage risk, e.g. To be followed...


Facility in original portfolio
Default
Censored
Begin/end study
Facility number
Time t(i) in years
0
1
2
3
4
5
6
7
-1
-0.5
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5

3 Pecularities of Machine Learning for finance

3. Pecularities of Machine Learning for finance

The Machine Learnine/AI/Big Data revolution has impacted all industries.

Main drivers:

- availability of huge datasets,
- computer storage and intensive calculations now possible,
- existence of models/algorithms/methods.

Here, the focus on the implications in finance (& insurance).

3. Pecularities of Machine Learning for finance

What is peculiar in finance, compared to other industries?

- time series with different scales (from daily to micro-seconds) and continuously updated (online learning);
- learning/simulation of market reactions/behaviors is difficult: low signal to noise ratio.
- large number of assets and clients $\Rightarrow$ clustering, PCA, segmentation, unsupervised learning.
- potentially complex optimization problems (portfolio optimization, investment strategies) $\Rightarrow$ reinforcement learning
- pricing/hedging under risk neutral (and not historical) probabilities. Etc.

What can ML techniques bring to existing methods?

New methods for

- classification (of clients, financial assets),
- regressions, conditional distributions (supervised learning),
- text/images/speech recognition $\Rightarrow$ the ability of exploiting new innovative data.

  Ex.: chats in Bloomberg flows, satellite images, news on the web, etc.

$\Rightarrow$ potentially a comparative advantage for the firms that understand/implement these new methodologies.

Consequences for many participants:

- asset managers: which portfolio will provide the best risk/return profile?
- market makers: what is the "right" price ? How can I optimally manage my inventory or hedge my book ?
- prop. traders, exotic traders: where are trading opportunities ?
- sales, financial advisors: which product (and when) will I propose to this client?
- risk managers: what is the risk of losses associated to this position/portfolio?
- compliance officers, operational risk executives: how can I identify fraud, money-laundering, mistakes, etc ?
- regulators, auditors: how will I deal with such a huge amount of information (quotes)? Are there some market manipulations?

3. Pecularities of Machine Learning for finance

Examples where the ML machinery may improve finance businesses:

(a) evaluation of financial risks: borrowers (loans, mortgages), firms (ratings), clients/counterparties.

$\Rightarrow$ "advanced" scoring methods

(b) building of investment strategies: stock picking, multi-assets portfolios,...
$\Rightarrow$ selection of factors, explanation of the equity risk premium, robot-advisors.

(c) pricing of derivatives by Monte-Carlo:
calculations/approximation of conditional expectations.

(d) Network modelling (of banking systems, counterparties, main dealers in a market, etc).

3. Pecularities of Machine Learning for finance

Q.: are "classical" models in statistics/econometrics dismissed by the emergence of ML methods ?

Property	Statistical inference	Supervised machine learning
Goal	Causal models with explanatory power	Prediction performance, often with limited explanatory power
Data	The data is generated by a model	The data generation process is unknown
Framework	Probabilistic	Algorithmic and Probabilistic
Expressibility	Typically linear	Non-linear
Model selection	Based on information criteria	Numerical optimization
Scalability	Limited to lower-dimensional data	Scales to high-dimensional input data
Robustness	Prone to over-fitting	Designed for out-of-sample performance
Diagnostics	Extensive	Limited

A large overlap between Machine Learning (ML) and "classical" statistical/econometric (SE) models.

Ex.: a linear model $Y = \mathbf{X}'\beta + \varepsilon$, $\mathbf{X} \in \mathbb{R}^p$, $\mathbb{E}[\varepsilon|\mathbf{X}] = 0$.

- $p << n$ and the law of ε is parametric $\rightarrow$ SE;
- $p \sim n \rightarrow$ ML;
- $p << n$ and the law of ε is left unspecified $\rightarrow$ SE or ML.

Roughly,

- traditional statistical/econometric models: try to "explain" the underlying process, with relatively few explanatory variables .
- Machine Learning models: focus on out-of-sample predictions, reduced-form models, highly parameterized.

Both approaches are often complementary.

Some methods belong to one domain or the other depending on the perspective/objectives.

Ex.: Can a deep neural network be considered as a non-parametric econometric model?

Ex.: many penalization methods are variations around usual linear models.

Essential types of financial data (Table 2.1 in [1])

Fundamental Data	Market Data	Analytics	Alternative Data
• Assets • Liabilities • Sales • Costs/earnings • Macro variables • ...	• Price/yield/implied volatility • Volume • Dividend/coupons • Open interest • Quotes/cancellations • Aggressor side • ...	• Analyst recommendations • Credit ratings • Earnings expectations • News sentiment • ...	• Satellite/CCTV images • Google searches • Twitter/chats • Metadata • ...

Different types of financial datasets

Basic information: set of transactions $i = 1, \ldots, N$.

For any transaction i:

- its calendar time t_i,
- its price p_i,
- its volume v_i (number of shares, securities),
- its value $c_i = p_i v_i$ in USD, Euro, or others.

Different types of financial datasets

Most of the time: aggregation of information

$\Rightarrow$ summaries of sets of transactions. But...many ways of doing this.

A model/trading strategy strongly depends on such choices!

Before doing anything on the modeling side, take your time for carefully deciding the nature of data that will be studied.

Different types of financial datasets

Sol. 1: "Time bars"

Consider all transactions recorded between two dates t_1 and t_2, and $[t_1, t_2]$ is a given fixed time interval: 1s, 5min, 1d, 1week, etc.

Available information:

- time stamp (t_1 or t_2)
- volume-weighted average price (VWAP)
- open/close/highest/lowest prices
- total traded volume,...

Different types of financial datasets

Sol. 2: "Tick bars"

Start from t_1 & do not fix t_2 ex ante.

Consider all transactions from t_1 until a fixed number N of transactions has been realized (at t_2).

No fixed time interval.

Available information: the same indicators as before, but calculated on a different subset of transactions.

Different types of financial datasets

Sol. 3: "volume bars"

Start from t_1 & do not fix t_2 ex ante.

Consider all transactions from t_1 until a total volume V_0 of transactions has been realized (at t_2).

No fixed time interval. *Available information:* the same indicators as before.

Sol. 4: "dollar bars" (or "euro bars")

Consider all transactions from t_1 until a total accumulated amount of cash C_0 has been exchanged (at t_2).

Different types of financial datasets

"Time bars": work/think in terms of calendar time.

"Tick/Volume/USD bars": work/think in terms of "trading time".

The i.i.d. assumption for the returns of financial series is false...
...but more acceptable in trading time than in calendar time:

Volatilities are a lot more stable for "volume bars" than for "time bars" (close to homoscedasticity).

Overlapping issues

Financial returns are not be i.i.d. in general.
Sometimes, they are related to overlapping periods of time.

This may be necessary... and may reduce explanatory powers.

Particular issues: bagging does not often work very well

Reminder: bagging = model averaging

$$\hat{Y}(x) = M^{-1} \sum_{j=1}^{M} f_j(x),$$

where the f_j are M predictors.

$\Rightarrow$ a smaller variance, in general.

A usual case: M bootstrap samples and predictors chosen in the same family of models/methods (c.f. random forests)

Overlapping issues

Every learner f_j is trained on a particular (randomly drawn) sub-sample $\mathcal{S}_j$.

The reduction in terms of variance is often poor when the dependence between successive returns is large.

More or less ad-hoc "solutions": block-bootstrap, diverse sequential bootstrap schemes, etc.

Boosting

Boosting are iterative algorithms to "boost" the predictive power of weak learners/predictors

Use of a reweighting scheme for observations

A boosting algorithm: Adaboost

1. Initialize the observation weights $w_i = 1/N,\ i = 1, 2, \ldots, N$.
2. For $m = 1$ to M:
   (a) Fit a classifier $G_m(x)$ to the training data using weights w_i.
   (b) Compute
   $$\mathrm{err}_m = \frac{\sum_{i=1}^{N} w_i I(y_i \neq G_m(x_i))}{\sum_{i=1}^{N} w_i}.$$
   (c) Compute $\alpha_m = \log((1 - \mathrm{err}_m)/\mathrm{err}_m)$.
   (d) Set $w_i \leftarrow w_i \cdot \exp[\alpha_m \cdot I(y_i \neq G_m(x_i))],\ i = 1, 2, \ldots, N$.
3. Output $G(x) = \mathrm{sign}\left[\sum_{m=1}^{M} \alpha_m G_m(x)\right]$.

Bagging vs Boosting

In Boosting:

- Individual classifiers are fit sequentially.
- Poor-performing classifiers are dismissed.
- Observations are weighted differently in each iteration.
- The ensemble forecast is a weighted average of the individual learners

Bagging vs Boosting

Boosting: reduces both variance and bias in forecasts, at the cost of greater risk of overfitting.

In financial time series applications bagging is generally preferable to boosting.

Bagging addresses overfitting, while boosting addresses underfitting.

Overfitting is often a greater concern than underfitting: not difficult to overfit a ML algorithm to financial data, because of the low signal-to-noise ratio.

Furthermore, bagging can be parallelized, while generally boosting requires sequential running.

Stationarity issues

For inference purpose, including ML techniques, the series have to be stationary.

Differentiate or not ? How many times ? What do we loose (memory) ?

Sol. 1: differentiation, but with a small order $\rightarrow$ fractional differentiation may be useful so that the initial series and the new one are as much as possible dependent

Sol 2: Co-integration

Structural breaks

Practical question: which history of quotes is relevant for my model?

Financial returns/variations are seldom stationary in the long term.

- structural breaks in the financial markets: changes of business models,
- mergers & acquisitions,
- new actors, new rules (index), new regulation, etc.

Intuition can fuel (ex.: before/after the financial crisis)...but a more rigorous approach is still possible.

CUSUM tests

Consider an assumed stationary time series $(x_t)_{t=1,\dots,T}$.

$\theta \in \mathbb{R}^m$ is a model parameter, as

- $\theta = \mathbb{E}[x_t]$
- $\theta = \mathbb{E}[x_t^2] - \mathbb{E}[x_t]^2$
- $\theta = Cov(x_t, x_{t-1})$, etc.

$\hat{\theta}_t$ an estimator of θ based on $x_1, \dots, x_t$. Assume

$$\sqrt{t}(\hat{\theta}_t - \theta) = \frac{1}{\sqrt{t}} \sum_{j=1}^{t} l_j + o(1),$$

where (l_t) is a stationary martingale difference: $\mathbb{E}[l_t | \mathcal{F}_{t-1}] = 0$.

CUSUM tests

We want to test

$$\mathcal{H}_0 : \theta \text{ does not change on any period of time } \{1, 2, \ldots, t\}.$$

Set $\Gamma := var(l_t)$.

$$T_n := \max_{m \leq t \leq T} \frac{t^2}{T} (\hat{\theta}_t - \hat{\theta}_T)' \hat{\Gamma}^{-1} (\hat{\theta}_t - \hat{\theta}_T)$$

is weakly convergent under $\mathcal{H}_0$.

Particular case: the mean of returns x_t, where $\theta = \mathbb{E}[x_t]$.

$\hat{\theta}_t - \theta = \frac{1}{t} \sum_{j=1}^{t} (x_j - \theta)$, $\Gamma = var(x_t)$

Ref.: Lee, Ha, Na, Na (2003). Scand. J. Statist.

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